ML System Design

A framework for design interviews

The problem

- Vague prompt: 'recommend videos', 'detect spam'
- Must cover data, model, serving and monitoring
- 45 minutes, so structure matters

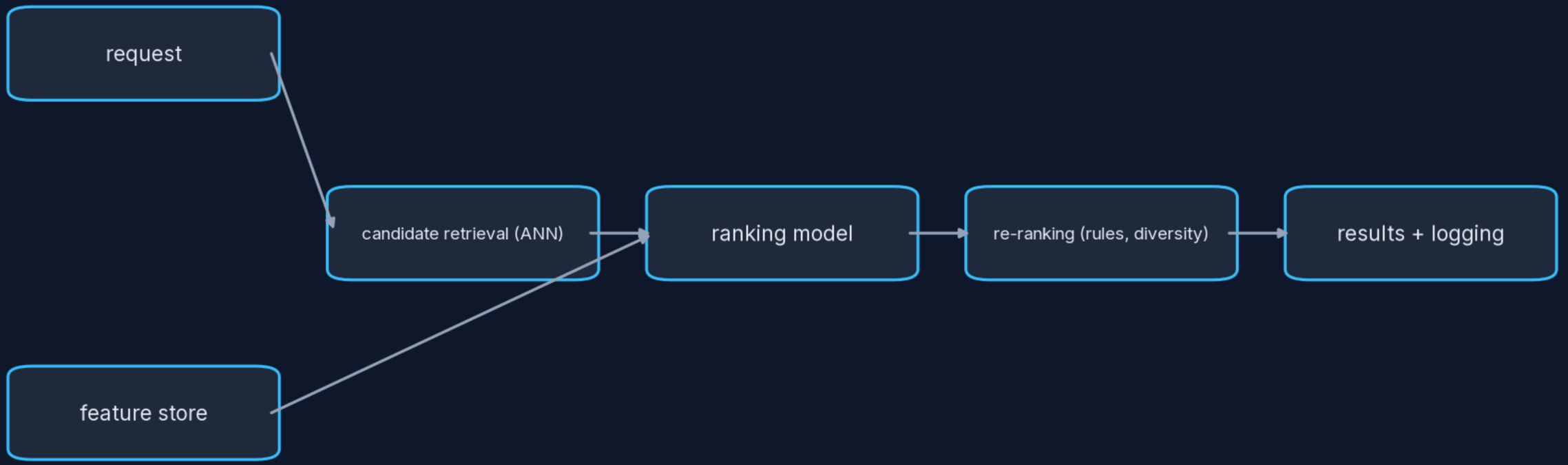
Framework, part 1

- 1. Clarify goal → ML objective
- 2. Metrics: offline (AUC, NDCG) + online (CTR, retention)
- 3. Data: labels, delay, imbalance, leakage, time splits
- 4. Features: user, item, context, embeddings, feature store

Framework, part 2

- 5. Model: simple baseline first, then deep
- 6. Training: loss, negative sampling, tuning
- 7. Serving: retrieval → ranking → re-ranking
- 8. Rollout: shadow, A/B test, canary
- 9. Monitoring: drift, freshness, retraining

Typical serving stack



Video recommendations

- Retrieval: two-tower model + ANN, plus subscriptions, trending
- Ranking: multi-task (click, watch time, like)
- Cold start: content embeddings, popularity, bandits

Spam, search, ads, ETA

- Spam: imbalanced → precision at fixed recall; review queue
- Search: BM25 + dense retrieval, LambdaMART, NDCG
- Ads: calibrated CTR (log loss), sparse features
- ETA: GBDT on segments; correct the routing estimate

Recap

- Goal → objective → metrics
- Data and features before models
- Baseline first; staged serving
- Roll out carefully; monitor and retrain